

GRANT UHER

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EDUCATION

Colorado School of Mines

B.S. Electrical Engineering (Pursuing Combined M.S. Track, May 2029); GPA: 4.0

Expected Dec 2028

Golden, Colorado

EXPERIENCE

Mines Formula SAE

Electrical Team Member

August 2025 – Present

- Led vehicle-level verification for the Brake System Plausibility Device (BSPD) and authored the Electronic Throttle Control safety documentation, securing full FSAE rule compliance.
- Collaborated with EV and IC Powertrain to define sensor integration, harness interfaces, and shutdown architecture.
- Designed BSPD shutdown circuitry to interface across both EV and IC platforms, incorporating inline shutdown switches across ignition coils, fuel injectors, and fuel pumps.
- Established the team's Altium workspace by building a shared component library, standardized layer stackups, and custom design rule checks to cut subsystem layout times for 15+ team engineers.
- Hand-soldered 0402 SMT components and executed PCB rework, including trace cuts and pin-level soldering on LQFP48 packages.

Computer Science Teaching Assistant

Colorado School of Mines

January 2026 – Present

Part-time

- Mentored students during weekly office hours, debugging code, clarifying core CS concepts, and accelerating student understanding across a 600+ student cohort.

PROJECTS

High-Speed Digital & GPS PCB

Altium Designer, STM32, GNSS, 100-BaseTX Ethernet

- Engineered a 100x increase in telemetry throughput (upgrading from 1Mbps CAN to a 100Mbps 100Base-TX), improving sensor sample potential.
- Designed a 4-layer microcontroller board centered on an STM32H563RIT6 MCU for high-speed data acquisition.
- Implemented the 100Base-TX Ethernet system, and integrated a high accuracy MAX-F10S GNSS module, designing an antenna power delivery network using a custom Bias Tee circuit.
- Routed 50Ω single-ended RMII and differential MDI pairs with controlled impedance stackups and continuous return paths to preserve high-speed signal integrity.

Brake System Plausibility Device (BSPD)

Altium Designer, LTSpice

- Engineered an entirely hardware-defined, non-programmable safety circuit using comparators, binary counters, and decade counters to trigger a time-delayed emergency shutdown signal.
- Developed a modular, dual-variant architecture using 0-ohm jumper configurations, allowing a single PCB design to seamlessly support both IC and EV vehicle rulesets.
- Designed an open-drain diagnostic architecture capable of isolating Loss of Signal (LOS) from simultaneous brake/throttle faults, doubling fault-detection capability.
- Implemented an op-amp signal conditioning stage to expand current sensor output resolution, validating circuit behavior through LTSpice simulations.

Solid-State Power Distribution & Relay Board

Altium Designer

- Engineered a power distribution PCB with high-side switches to manage critical engine subsystems, routing power planes and optimizing thermal limits to support continuous 7A loads.
- Designed a custom bi-directional MOSFET solid-state relay to interrupt an H-bridge driven servo.

TECHNICAL SKILLS

Hardware & EDA: Altium Designer, LTSpice, Verilog, High-Speed Routing, PCB Rework, SMT Assembly (0402, LQFP)

Protocols & Systems: CAN, RMII, MDI (100Base-TX Ethernet), SPI, I2C, USB

Software & Tools: Python, C++, SolidWorks (CSWA Certified), Fusion 360

Lab & Testing: Oscilloscopes, Digital Multimeters (DMM), Function Generators, Power Supplies, Hot Air Rework Stations, Soldering Iron